

ADNAN ABBAS

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Research Interests

- Designing agentic systems that act on external environments and maintain context across multi-turn conversations.
- Understanding how people negotiate and calibrate the autonomy of goal-driven agents for behavior change.
- Studying long-term human–AI adaptation and its implications for collaboration, trust, and knowledge work practices.

Education

Virginia Tech

Ph.D. in Computer Science (HCI) – Advised by Dr. Sang Won Lee

Aug 2023 – Present

Blacksburg, VA, USA

Lahore University of Management Sciences (LUMS)

Bachelor of Science in Computer Science

Aug 2018 – May 2022

Lahore, Pakistan

Publications

Journal & Conference Proceedings

- Sangwook Lee, [Adnan Abbas](#), Sang Won Lee, Young-Ho Kim, Yan Chen, “*MAESTRO: Adapting GUIs and Guiding Navigation with User Preferences in Conversational Agents with GUIs*,” in *Proceedings of the 39th Annual ACM Symposium on User Interface Software and Technology (UIST 2026)*.
- [Adnan Abbas](#), Caleb Wohn, Arnav Jagtap, Eugenia H. Rho, Young-ho Kim, Sang Won Lee, “*‘Having Lunch Now’: Understanding How Users Engage with a Proactive Agent for Daily Planning and Self-Reflection*,” in *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2026)*.
- Sangwook Lee, [Adnan Abbas](#), Yan Chen, Young-ho Kim, Sang Won Lee, “*CHOIR: A Chatbot-mediated Organizational Memory Leveraging Communication in University Research Labs*,” in *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2026)* **Honorable Mention Award (Top 5% of submissions)**.
- [Adnan Abbas](#), Caleb Wohn, Donghan Hu, Eugenia H. Rho, Sang Won Lee, “*PITCH: Designing Agentic Conversational Support for Planning and Self-reflection*,” in *Proceedings of the ACM Conference on Conversational User Interfaces (CUI 2025)*.
- Mukhtiar Ahmad, Syed Muhammad Nawazish Ali, Muhammad Taimoor Tariq, Syed Usman Jafri, [Adnan Abbas](#), et al., “*Neutrino: A Fast and Consistent Edge-based Cellular Control Plane*,” in *IEEE/ACM Transactions on Networking*, 2023.

Papers Under Review

- [Adnan Abbas](#), Linda W. Chang, Lu Sun, Laura A. Libby, “*BRIDGE: Proactive Human–AI Collaboration for Behavior-Change Design*,” manuscript under review for the *ACM Conference on Human Factors in Computing Systems (CHI 2027)*.

Lightly Reviewed Papers (Workshops & Posters)

- Sangwook Lee, [Adnan Abbas](#), Yan Chen, Sang Won Lee, “*CHOIR: Chat-based Helper for Organizational Intelligence Repository*,” in *ACM CHI 2025 Workshop on Tools for Thought*.
- [Adnan Abbas](#), Sang Won Lee, “*PITCH: Productivity and Mental Well-being Coaching through Daily Conversational Interaction*,” in *ACM CHI 2024 Workshop on Designing AI for Well-being*, Honolulu, HI.

Research Methods and Skills

Research Methods: Mixed-methods design, field deployment, semi-structured interviews, storyboarding, thematic analysis, dialogue act coding, statistical modeling

Technical Skills: Python, JavaScript (Node.js, React), C++, Java, SQL, HTML/CSS, Bash

Frameworks & Tools: OpenAI API, Flask, MongoDB, MERN stack, Slack Bolt, Docker, GitHub, Figma, Balsamiq, Nginx

Work Experience

Graduate Research Assistant - Echolab

Virginia Tech

Aug 2023 – Present

Blacksburg, VA, USA

- Independently drive research from concept to publication, drawing on human-AI interaction, behavioral science, and NLP literature to develop human-in-the-loop agentic systems.
- Collaborate with psychology and social science researchers to design AI agents that support executive functioning, planning, and self-reflection.
- Design and conduct mixed-methods field studies that combine qualitative interviews with quantitative measures of user behavior and self-reported outcomes to evaluate systems and inform iterative design.
- Analyze interviews and interaction data using thematic analysis and statistical methods in R, including regression, generalized linear mixed models (GLMMs) to translate findings into actionable design recommendations.
- Present research findings to interdisciplinary audiences and write and prepare manuscripts for peer-reviewed publication.

Research Intern

Toyota Research Institute

May 2026 – Sep 2026

- Created an AI tool for behavioral scientists to accelerate the process of diagnosing behavior problems and designing behavior change interventions
- Designed and implemented a proactive conversational AI prototype combining agent orchestration and tool calling.
- Coordinated IRB approval and designed and conducted a counterbalanced within-subjects experiment (N = 16).
- Combined thematic analysis with mixed-effects modeling in R to examine quantitative metrics, identifying design tradeoffs between proactive guidance and user agency.
- Presented research progress and findings to stakeholders and prepared a full-paper submission to CHI 2027.

Educational Technology Engineer

Educative, Inc.

Jul 2022 – Jul 2023

Lahore, Pakistan

- Performed a literature review on spatial computing and VR teaching solutions; presented findings that informed leadership's decision to expand into XR ed-tech
- Led the development of flagship Web 3.0 and XR courses, [WebXR Experiences](#) and [VR Development with Unity](#)
- Collaborated with PMs and developers to engineer a novel WebADB-based VR widget, enabling in-browser headset integration and reducing configuration overhead for learners
- Conducted usability tests (N=20) uncovering pain points in navigation and learner comprehension

Research Assistant

Lahore University of Management Sciences

Jul 2020 – May 2022

Lahore, Pakistan

- Designed and developed a novel system design for optimizing control plane operations in edge-based cellular networks
- Developed a user state management algorithm leveraging geographically aware consistent hashing
- Designed test cases and improved the performance of latency sensitive applications (AR/VR, self-driving cars) by 2.5x and 2.8x, under real-world cellular traffic

Projects

Understanding Organizational Memory Practices | Interview Study, Design Feedback

July 2025

- Conducted formative interviews (N=15) with faculty to understand organizational memory challenges & practices
- Synthesized findings into design goals ([Demo Video](#))

Encouraging Reflection in Journal Writing | Human-LLM Interaction, Prototype Development

May 2024

- Designed and developed an LLM-based journaling prototype enabling users to set future questions and control topic focus through text interactions
- Implemented guided prompts and evolving personalization features to support reflective writing ([Demo Video](#))

Short-form Content Addiction in Virtual Reality | Within-subject experiment, Quantitative Evaluation

Dec 2023

- Conducted a within-subject user study (N=15) comparing short-form content use in VR vs. smartphones through the lens of flow theory, examining engagement, time distortion, and well-being
- Analyzed quantitative findings and authored the report [SFC Addiction in VR Through the Lens of Flow Theory](#).

SkillCoach: Connecting Learners and Tutors | Figma Prototyping, Mixed-Methods Evaluation

Oct 2022

- Designed a peer-to-peer skill-sharing mobile platform connecting learners with tutors ([Case Study](#), [Figma Prototype](#))

Academic Involvement

Teaching

- User Interface Software (CS 5774) Fall 2026
Graduate Teaching Assistant, Virginia Tech Blacksburg, VA
- Database Management Systems (CS 5614) Spring 2026
Graduate Teaching Assistant, Virginia Tech Blacksburg, VA
- Ethics & Professionalism in Computer Science (CS 5024) Summer 2025
Graduate Teaching Assistant, Virginia Tech Blacksburg, VA
- Introduction to GUI Programming / Graphics (CS 3744) Spring 2025
Graduate Teaching Assistant, Virginia Tech Blacksburg, VA
- Software Design and Data Structures (CS 2114) Fall 2023 – Fall 2024
Graduate Teaching Assistant, Virginia Tech Blacksburg, VA
- Network-Centric Computing (CS 382) Fall 2021
Undergraduate Teaching Assistant, LUMS Lahore, Pakistan

Reviewer

- CHI 2026 Oct 2025
Recognized for outstanding review

Student Volunteer

- CHI 2026 May 2026

Mentorship

- Benjamin Wheatley Jun 2025 – Present
Undergraduate Research Assistant, Virginia Tech Blacksburg, VA
- Arnav Jagtap Jan 2025 – May 2025
Undergraduate Research Assistant, Virginia Tech Blacksburg, VA
- Amanjeet Sahagal Jun 2025 – Sep 2025
Undergraduate Research Assistant, Virginia Tech Blacksburg, VA